

Coarctation of the Aorta



Section I: Scenario Demographics

Scenario Title:	Congenital Heart Disease		
Date of Development:	25/02/2016 (DD/MM/YYYY)		
Target Learning Group:	☐ Juniors (PGY 1 – 2)	☒ Seniors (PGY ≥ 3)	☐ All Groups

Section II: Scenario Developers

Scenario Developer(s):	Dr. Quang Ngo & Dr. Donika Orlich
Affiliations/Institution(s):	McMaster University
Contact E-mail (optional):	Donika.orlich@medportal.ca

Section III: Curriculum Integration

Learning Goals & Objectives	
Educational Goal:	To expose learners to neonatal shock due to severe coarctation of the aorta.
CRM Objectives:	1) Demonstrates good problem solving including quick implementation of concurrent management plans 2) Demonstrates situational awareness by avoiding fixation error and anticipating likely events
Medical Objectives:	1) Employs a structured approach to the investigation and management of cyanosis in a neonate 2) Initiates treatment for likely ductal dependent cardiac lesion 3) Anticipates the complications of PGE infusion and implements a rational approach to intubation in the neonate

Case Summary: Brief Summary of Case Progression and Major Events

A 2 week old neonate presents in shock requiring the learner to implement an initial broad work-up. The patient will also be hypoglycemic, and will seize if this is not promptly recognized. Physical exam and CXR findings will suggest coarctation of the aorta as the likely cause, and the learner should recognize the need for gentle fluid boluses and a PGE infusion. The infant will become apneic after starting the PGE infusion, requiring intubation.

References

- Marx, J. A., Hockberger, R. S., Walls, R. M., & Adams, J. (2013). *Rosen's emergency medicine: Concepts and clinical practice*. St. Louis: Mosby.
- Al Balushi A, Zacharias S, Al Senaidi K. Coarctation of the Aorta, Known Yet Can Be Missed. Oman Med J 2013 May; 28(3):204-206. Available from http://www.omjournal.org/fulltext_PDF.aspx?DetailsID=380&type=fulltext



Coarctation of the Aorta



Section IV: Scenario Script

A. Scenario Cast & Realism			
Patient:	☒ Neonatal Computerized Mannequin ☐ Mannequin ☐ Standardized Patient ☐ Hybrid ☐ Task Trainer	Realism: <i>Select most important dimension(s)</i>	☒ Conceptual ☐ Physical ☒ Emotional/Experiential ☐ Other: ☐ N/A
Confederates	Brief Description of Role		
Mother:	Brought patient to the ED as she was worried because her son hasn't been feeding very well. After vomiting the last feed, she noticed he was quite lethargic and seemed a strange colour. ROS: no fever, no diarrhea, no sick contacts, no urine since last night PMHx: mother G3P3; uncomplicated pregnancy, GBS neg, delivered SVD at 38 weeks; no meconium, no PROM, no birth complications and left on day 2 of life, breastfeeding		
B. Required Monitors			
☒ EKG Leads/Wires	☐ Temperature Probe	☐ Central Venous Line	
☒ NIBP Cuff	☐ Defibrillator Pads	☐ Capnography	
☒ Pulse Oximeter	☐ Arterial Line	☐ Other:	
C. Required Equipment			
☒ Gloves	☒ Nasal Prongs	☐ Scalpel	
☒ Stethoscope	☒ Venturi Mask	☐ Tube Thoracostomy Kit	
☐ Defibrillator	☒ Non-Rebreather Mask	☐ Cricothyroidotomy Kit	
☒ IV Bags/Lines	☒ Bag Valve Mask	☐ Thoracotomy Kit	
☒ IV Push Medications	☒ Laryngoscope	☐ Central Line Kit	
☐ PO Tabs	☐ Video Assisted Laryngoscope	☐ Arterial Line Kit	
☐ Blood Products	☒ ET Tubes	☐ Other:	
☒ Intraosseous Set-up	☐ LMA	☐ Other:	
D. Moulage			
None			
E. Approximate Timing			
Set-Up:	5 min	Scenario:	20 min
		Debriefing:	40 min

Coarctation of the Aorta



Section V: Patient Data and Baseline State

A. Clinical Vignette: To Read Aloud at Beginning of Case

Your triage nurse comes to tell you about an infant she just put in the resuscitation room who she feels looks quite unwell. Sam is a 2 week old neonate brought to the ED by his mother. She was worried because he hasn't been feeding very well. He seems to just get sleepy when feeding. Now he just vomited his last feed and seems really lethargic. She thinks he just "doesn't look the right colour".

B. Patient Profile and History

Patient Name: Sam Lodger Age: 2 weeks Weight: 3kg

Gender: ☒ M ☐ F Code Status: Full Code

Chief Complaint: Vomiting

History of Presenting Illness: As above by mother.

Past Medical History:	None	Medications:	None
	Born 38 weeks GA (SVD)		

Allergies: None

Social History: Lives with mom and dad. 2 Older siblings at home.

Family History: No hx CHD, inborn errors of metabolism, endocrine disorders. Mother healthy

Review of Systems:	CNS:	Mom feels he is much more lethargic than usual.		
	HEENT:	Nil		
	CVS:	Nil		
	RESP:	Nil		
	GI:	Poor feeding. Vomited the last feed. No diarrhea. No hematochezia.		
	GU:	No wet diaper since last night		
	MSK:	Nil	INT:	Cool skin. ? normal color

C. Baseline Simulator State and Physical Exam

☐ No Monitor Display ☒ Monitor On, no data displayed ☐ Monitor on Standard Display

HR: 180/min BP: 80/40 (R arm) RR: 50/min O₂SAT: 94 %

Rhythm: NSR T: 36.0°C Glucose: 1.5mmol/L GCS: 12 (E2 V4 M6)

General Status: Grunting and very lethargic infant.

CNS: Floppy and lethargic with little response. Moving limbs symmetrically

HEENT: PERLA 2mm. No signs HI. Fontanelle sunken.

CVS: Normal S1, S2. No murmurs. Weak lower limb and femoral pulses, normal upper limb pulses.
4 limb BPs – RA 80/40 LA 82/44 RL 65/35 LL 67/40

RESP: Crackles to the bases. No increased work of breathing. Tachypneic.
O₂ Sats- RA 94% LL 89%

ABDO: Soft. No distension. No apparent discomfort.

GU: Nil

MSK: No signs abuse. SKIN: Delayed CRT 4 sec and grey appearance

Coarctation of the Aorta



Section VI: Scenario Progression

Scenario States, Modifiers and Triggers

Patient State	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State	
1. Baseline State Rhythm: NSR HR: 180/min BP: 80/40 RR: 50/min O ₂ SAT: 94% T: 36°C **BP cuff and O₂ probe both start on right arm. Unless asked for 4 limb BP's, will not move cuff to check other limbs.	Lethargic. Opens eyes to pain. Moves spontaneously. Weak moaning. Looks grey with delayed cap refill. Some peripheral cyanosis.	Learner Actions - ☐ Monitors, insert iv - ☐ 100% O ₂ - ☐ Cap Glucose (1.5mmol/L) - ☐ D10 bolus (5cc/kg) for hypoglycemia - ☐ IV NS bolus (10cc/kg) - ☐ Broad Antibiotics (Ampicillin + Cefotaxime) - ☐ Labs (basic + lactate + VBG + ammonia + LFTs + BC) - ☐ Check 4 limb BP's - ☐ CXR - ☐ ECG - ☐ Early consult of PICU, RT - ☐ Delegate team member to take history from mother & update her	Modifiers <i>Changes to patient condition based on learner action</i> - 100% O ₂ → no improvement in distal cyanosis Triggers <i>For progression to next state</i> - Hypoglycemia not recognized by 3 mins → 2. Seizure - 5 mins → 3. Hypotension
2. Seizure Rhythm: NSR HR: 190/min BP: 90/50 RR: 10/min O ₂ SAT: 97%	Patient begins to seize	Learner Actions - ☐ Cap glucose (1.5) - ☐ D10 bolus (5cc/kg)	Modifiers - IV benzo → continues to seize - D10 bolus → seizure stops Triggers - Hypoglycemia corrected → 3. Hypotension
3. Hypotension Rhythm: NSR HR: 190/min BP: 55/40 legs, 65/35 arms RR: 25/min O ₂ SAT: 98%	More lethargic. Eyes open only to pain. No verbal response. Flexion to pain. Cap refill now 6s (more grey)	Learner Actions - ☐ IV NS bolus (10cc/kg) x3 - ☐ Anticipate need for Dopamine (5mcg/kg/min) OR Epi (0.05mcg/kg/min) - ☐ Plan for intubation (anticipate apnea with PGE1) - ☐ Chose correct tube size (3.5 uncuffed, 3.0 cuffed) - ☐ ± Pretreat with atropine (0.2mg/kg) - ☐ Etomidate ideal agent - ☐ Paralytic - ☐ Post-intubation sedation (midazolam) - ☐ Start PGE 1 infusion (0.05-0.1mcg/kg/min)	Modifiers - Nurse to prompt re: 4 limb BP's if not yet done - 1 st NS bolus → HR 185, BP 58/40 - 2 nd bolus → HR 180, BP 60/45 - 3 rd bolus → HR 170, BP 65/48 Triggers - Intubation without atropine → 4. Bradycardia - Intubation + PGE1 started → 7. Resolution - PGE1 without intubation → 5. Apnea - No PGE1 started by 5 mins → 6. Arrest

Coarctation of the Aorta



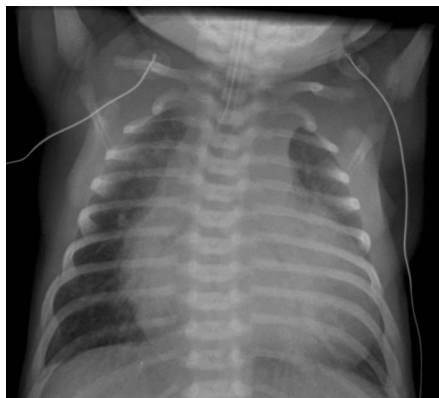
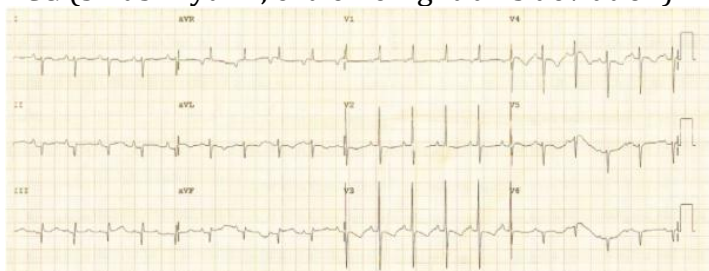
4. Bradycardia O ₂ SAT → 95% over 1 min of bagging HR → 70 over 1 min BP 40/20	Baby becomes less responsive.	<u>Learner Actions</u> - ☐ Give atropine 0.02mg/kg iv (0.56mg) - ☐ Plan for intubation (as above) - ☐ Start PGE 1 infusion (0.05-0.1mcg/kg/min) if not already given	<u>Modifiers</u> - Atropine given → HR 160, O ₂ 98% <u>Triggers</u> - No atropine given by 3 mins → 6. Arrest - No PGE1 started by 5 mins → 6. Arrest - Intubated and PGE1 started → 7. Resolution
5. Apnea O ₂ SAT → 90% over 1 min of bagging RR → 3 over 1 min HR: 170 BP: 55/40	Baby stops breathing	<u>Learner Actions</u> - ☐ Start BMV ventilation - ☐ Plan for intubation (as above)	<u>Modifiers</u> - If not considering intubation, slowly decrease O ₂ stats → 85% <u>Triggers</u> - Intubation → 7. Resolution
6. Arrest HR: 35 BP: 40/20 O ₂ : 92% NRB/BMV RR: 8 ETCO ₂ : 10	Baby becomes unresponsive	<u>Learner Actions</u> - ☐ BMV ventilate - ☐ Start chest compressions (depth = ½ AP diameter of chest) - ☐ 15:2 compressions to breaths (8 breaths/min if intubated) - ☐ Recheck glucose (5.2) - ☐ IV NS bolus (10cc/kg) - ☐ Epinephrine (0.01mg/kg q3min) - ☐ Plan for intubation (as above) - ☐ Atropine (0.02mg/kg) if from state 4. Bradycardia - ☐ Start PGE 1 infusion (0.05-0.1mcg/kg/min) if not already started	<u>Modifiers</u> - BMV ventilation → no change - Chest Compressions → ETCO ₂ 20 <u>Triggers:</u> - If cause of arrest no PGE1, once PGE1 started + intubation → 7. Resolution - If cause of arrest no atropine, once atropine given + intubated → 7. Resolution
7. Resolution HR: 150/min BP: 80/40 RR: 20/min (vented) O ₂ SAT: 99% T: 36°C	Patient intubated and sedated	<u>Learner Actions</u> - ☐ Post-intubation CXR - ☐ Update ICU - ☐ Consult Cardiac Sx - ☐ Update mother	<u>Triggers</u> - 20 mins or all actions complete → END CASE

Coarctation of the Aorta



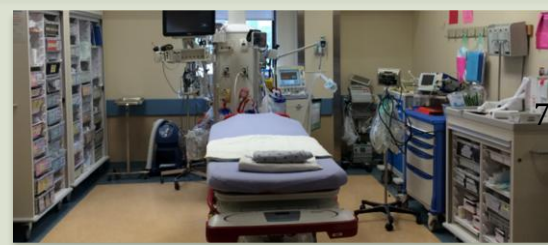
Section VII: Supporting Documents, Laboratory Results, & Multimedia

Laboratory Results						
Na: 140	K: 5.6	Cl: 105	HCO ₃ : 25	BUN: 5	Cr: 80	Glu: 1.5
Ca: 2.50	Mg: 0.8	PO ₄ :	Albumin: 35			
VBG	pH: 7.29	PCO ₂ : 50	PO ₂ : 95	HCO ₃ : 18	Lactate: 3.0	
WBC: 18	Hg: 140	Hct: 0.4	Plt: 550			

Images (ECGs, CXRs, etc.)	
Initial CXR (CHF, cardiomegaly)  CXR source: http://www.adhb.govt.nz/newborn/TeachingResources/Radiology/CXR/HLHS/CXR-HLHS-congested.jpg	Post-intubation CXR  CXR source: http://www.adhb.govt.nz/newborn/TeachingResources/Radiology/CXR/OtherCHF/NonstructuralCHF.jpg
ECG (sinus rhythm, extreme right-axis deviation)  ECG source: http://www.omjournal.org/IssueText.aspx?issId=380	
Ultrasound Video Files (if applicable)	
None	



Coarctation of the Aorta



Section VIII: Debriefing Guide

General Debriefing Plan	
☐ Individual	☒ Group
☐ With Video	☒ Without Video
Objectives	
Educational Goal:	To expose learners to neonatal shock due to severe coarctation of the aorta.
CRM Objectives:	1) Demonstrate good problem solving including quick implementation of concurrent management plans considering alternatives during crisis 2) Demonstrates situational awareness, avoiding fixation error, anticipating likely events, and constantly reassessing the situation 3) Demonstrates leadership skill by staying calm during crisis to maintain a global perspective and demonstrate prompt decision making
Medical Objectives:	1) Employs a structured approach to the investigation and management of cyanosis in a neonate 2) Initiates treatment for likely ductal dependent cardiac lesion 3) Anticipates the complications of PGE infusion and implements a rational approach to intubation in the neonate
Sample Questions for Debriefing	
1) What is your initial DDx for neonatal shock? What antidotes/treatments can be initiated immediately in these patients? 2) What was your approach to hypotension for this patient? How might your approach to fluid resuscitation change in a patient in whom CHD is likely? 3) What features on Hx, PE, CXR and ECG suggested the diagnosis of coarctation of the aorta? 4) What are the ductal dependent lesions? How are these managed in the ED? 5) What are common complications of prostaglandin infusion? 6) What are the clinical features of congenital heart disease? When should it be suspected in infants presenting to the ED? How was this patient the same? How was this patient different?	
Key Moments	
1) Recognition of likely cardiac cause to cyanosis and initiation of PGE1 infusion for possible ductal dependent lesion 2) Tailored approach to fluid resuscitation and vasopressor choice given likely CHD 3) Anticipating need for intubation given apnea as a common side effect to PGE1 infusion	